

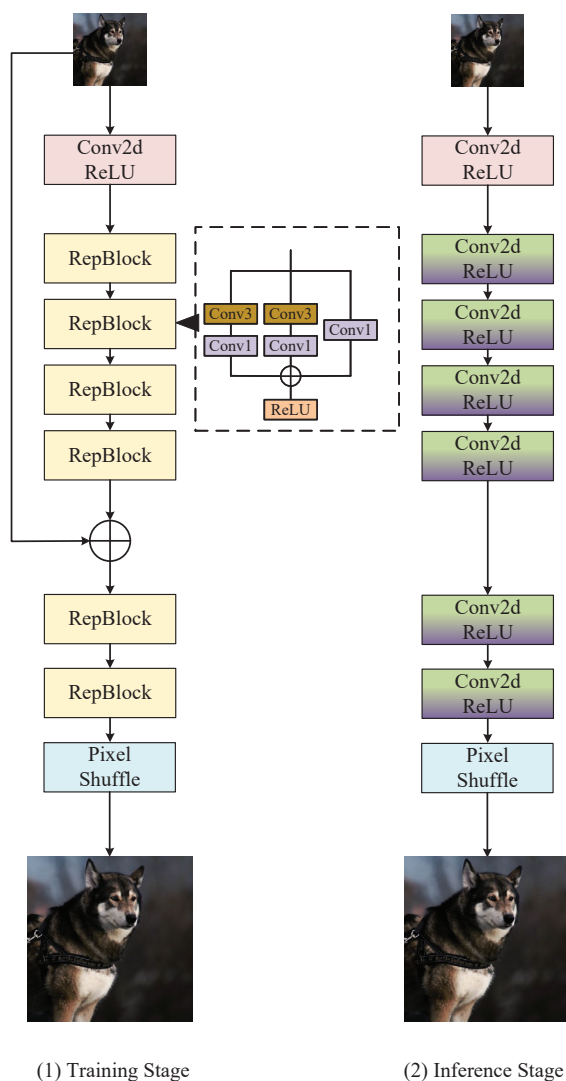


Figure 10. CNN architecture proposed by team HFUT LVGroup

sequence. Subsequently, two more RepBlocks process the features before a final Pixel Shuffle layer performs upsampling. The RepBlocks used parallel convolutions ( $3 \times 3$  and  $1 \times 1$ ) which are summed and then passed through ReLU. During the inference stage, these RepBlocks are fused into simple Conv  $3 \times 3$  + ReLU layers, and the residual skip connection is removed, yielding an efficient sequential network.

The model was trained on the DF2K dataset (comprising DIV2K and Flickr2K) using a three-stage, iteration-based approach. Stage one involved 1 million iterations using  $L_1$  loss and the AdamW optimizer, with an initial learning rate of  $5 \times 10^{-4}$  halved every 200K iterations. A batch size of 32 and  $48 \times 48$  low-resolution patches were used. Stage

two continued for 500K iterations, employing a combination of  $L_2$  loss and a block distillation loss, with the learning rate reduced to  $5 \times 10^{-5}$ . The third and final stage ran for 200K iterations, fine-tuning the model with the addition of an  $0.1 \cdot FFT$  loss to the existing loss terms. The learning rate for this stage was  $5 \times 10^{-6}$ , and the input patch size was increased to  $96 \times 96$ . The AdamW optimizer and a batch size of 32 were likely maintained throughout stages two and three.

## 5. Additional Literature

An overview of the past challenges on mobile-related tasks together with the proposed solutions can be found in the following papers:

- Image Super-Resolution: [9, 33, 68, 83]
- Learned End-to-End ISP: [35, 39, 47, 48, 89]
- RGB Image Enhancement: [29–31, 33]
- Bokeh Effect Rendering: [34, 37, 38]
- Image Denoising: [1, 2, 22]

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## A. Teams and Affiliations

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### AntSR

#### Title:

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